MIL-B-60073A(MU)

5 August 1966
SUPERSEDING
MIL-B-60073(MU)
18 June 1964

MILITARY SPECIFICATION

BOMB, NONPERSISTENT GAS, SPHERICAL, 4.5 INCH, M139; FILLING AND ASSEMBLY

1. SCOPE

1.1 This specification covers the filling and assembling of one type of spherical bomb.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-W-8604 - Welding of Aluminum Alloys: Process For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY EDGEWOOD ARSENAL

DL14-5-2768 - Bomb, Nonpersistent Gas, Spherical, 4.5 Inch, M139 - Filling and Assembly.

(All drawings referenced in this specification are listed on above drawing.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1325

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3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain prior to their use that all materials and components have met their respective specifications.

3.2 Welding procedure. Prior to the start of any production, the supplier shall establish an approved welding procedure in accordance with MIL-W-8604 except that the paragraphs describing maintenance sampling and cleaning are not applicable. The welding procedure shall include provisions for repair and rework of the weldment. The supplier shall establish a quality control procedure to be used during all production to assure that the approved welding procedure is maintained and controlled, and is producing welds conforming to MIL-W-8604. The supplier shall requalify his procedure whenever it varies from the accepted welding procedure. Compliance for this requirement shall be in accordance with 4.3.4.4.

3.3 Filling. The shell halves shall be filled with a constant volume of chemical agent weighing 288 to 295 grams as determined by net weight based on a calibration of volume of the shell half and verified in accordance with 4.3.4.3.

3.4 Closing. The bomb shell halves shall be closed as shown on Drawing C14-5-2768. From the time of filling until completion of the test in accordance with 4.3.4.2, the shell halves shall be kept in a closure-up position (see 6.4). Only one repair weld is permitted for any welded closure.

3.5 Agent contamination. The exposed surfaces of the bomb shall be free from GB contamination (see 4.3.4.1 and 6.5).

3.6 Helium leakage. The bomb shell halves shall not leak helium through the closure at a rate greater than 1×10^{-6} atmospheric cubic centimeter (cc) per second when tested in accordance with 4.3.4.2.

3.7 Assembly. The bomb shall be assembled as shown on Drawing C14-5-2768.

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of bombs shall be produced in accordance with this specification for examination and test (see 4.2).

3.9 Workmanship. The bombs shall be free from foreign matter and visible damage such as dents and cracks which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as

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specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Lotting. Prior to the start of production, a preproduction lot of 200 bombs shall be produced using the same methods, materials, procedures, and equipment that will be used in regular production.

4.2.2 Examination and test. Each bomb shall be examined and tested for all the requirements listed in this specification.

4.2.3 Acceptance/rejection criteria. The preproduction lot shall meet all the requirements listed in this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the bombs filled by one manufacturer, on one filling line under essentially the same filling conditions. However, no more than two lots of boosters, bursters and fuzes and no more than one lot of agent shall be represented in any one lot of filled and assembled bombs.

4.3.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection provisions.

4.3.3.1 Examination. Sample bombs shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.3.3.2 For critical defects. Each bomb in the lot shall be inspected for its critical characteristics listed in the classification of defects.

4.3.3.3 Classification of defects.

(a) Bomb, nonpersistent gas, spherical, 4.5 inch, M139 - assembly (Dwg C14-5-2768).

Categories	Defects	Acceptance standards
Critical:		
1	Co nta mination	4.3.4.1
2	Helium leakage	4.3.4.2
3	Filling excessive	4.3.4.3

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4.3.3.3 Classification of defects. (continued)

(a) Bomb, nonpersistent gas, spehical, 4.5 inch, M139 - assembly (Dwg C14-5-2768).

Categories	Defects	Acceptance standards
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Fastener loose after crimping	
103	Booster, burster or fuze loose after assembly	
104	Marking incorrect or illegible	
105	Workmanship (see 3.9)	
Minor:	AQL 2.5 percent defective	
201	Filling insufficient	

4.3.4 Tests.

4.3.4.1 GB contamination. Contamination shall be determined by the use of an approved Edgewood Arsenal GB agent detector system.

4.3.4.2 Helium leakage. Prior to closing, the void above the agent shall be evacuated to a pressure no greater than 10.2 pounds per square inch absolute (psia) and filled with helium to atmospheric pressure. The closure area shall be sealed off by a test head manifold to a vacuum system and a mass spectrometer type helium leak detector. The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during the test. The leak detector shall be calibrated to respond to a leak of 1×10^{-6} atmospheric cc per seoond of helium at the test head when a standard leak is introduced into the system at that point.

4.3.4.3 Filling level. The quantity of GB in the filled shell half shall be determined by weighing the shell half including the plug prior to filling and then after filling and closing. The scale shall be accurate to $\pm$ 0.1 gram (see 6.3 and 6.4).

4.3.4.4 Welding procedure. The supplier shall produce a qualification lot of 50 bombs using the proposed welding and control procedures to demonstrate the satisfactory quality of the weld produced. Evaluation and approval shall be based upon the following:

(1) Review of written procedures (including cleaning and timing sequence).
(2) Microscopic and macroscopic examination for:

(a) Fusion and joint penetration.
(b) Cracks.

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(c) Porosity and inclusions.
(d) Overlapping.
(e) Other metallic discontinuities.

(3) Helium test of closure.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking information is not applicable to this specification.

6. NOTES

6.1 Intended use. This item is used as the agent container for chemical warheads.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

1. Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
2. Name and address of test facility and shipping instructions when testing is performed by the Government.
3. Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Filling weight. When filled as specified in 3.3, the shell half carries approximately 0.6 pound of chemical agent. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Precautions. In view of the hazards attending in handling and filling of munitions with GB, suppliers are advised to exercise all necessary precautions to protect personnel from injury during the filling and handling. Care should be taken to prevent wetting the closure with agent thereby making void the results of the helium leak test.

6.5 Decontamination. Decontamination can be accomplished by subjecting the bomb to a spray of decontaminating solution (5- to 10-percent commercial grade soda ash, Na_2CO_3) for a minimum period of 1 minute at a minimum pressure of 15 psig in such a manner as to completely wash all surfaces. The bomb should then be rinsed with water to remove all traces of the decontaminating solution. The temperature of the spray and rinse water should be maintained at a range of 140° to 160°F. Spray pressures should have gages with an accuracy of 1 psig. An approved procedure of control of the decontaminating solution strength should be maintained by periodic examination and the time the bomb is subjected to the spray should be automatically controlled and recorded.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1325-A166

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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